23	C	Capacitance for each parallel network = C + C + C = 3C Total capacitance = $(1 / 3C + 1 / 3C)^{-1} = 3C / 2$
24	B	P and Q have the same speed (pass through velocity selector undeviated). Inside uniform field, both ions travel in circular paths. Magnetic force provides the centripetal force: $BQv = \frac{mv^2}{r}$ $\Rightarrow r = \frac{mv}{QB}$ Since B is constant and P & Q have the same speed, we have $r \propto m / Q$. So, $r_Q = r_P \left(\frac{m_Q}{m_P}\right)\left(\frac{Q_P}{Q_Q}\right) = 3.7\left(\frac{1.5}{1}\right)\left(\frac{1}{2}\right) = 2.8$ cm
25	B	Magnetic flux density $B = F/IL$ Max induced e.m.f. when speed of wire is the max